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United States Patent	12406565
Kind Code	B2
Date of Patent	September 02, 2025
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Aerosol generation device locator

Abstract

An attachable accessory for device location of an aerosol generation device may include a sleeve portion, a listening module and an alerting module. The sleeve portion may be configured to engage a portion of the aerosol generation device to operably couple the accessory to the aerosol generation device. The listening module may be configured to receive an alert signal from a user. The alerting module may be configured to issue an alert responsive to receipt of the alert signal by the listening module.

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Appl. No.: 17/380574

Filed: July 20, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220028245 A1	Jan. 27, 2022

Related U.S. Application Data

us-provisional-application US 63054452 20200721

Publication Classification

Int. Cl.: G08B21/24 (20060101); **A24F40/60** (20200101); **A24F40/65** (20200101); **G08B3/10** (20060101); **H04W4/029** (20180101)

U.S. Cl.:

CPC G08B21/24 (20130101); **A24F40/60** (20200101); **A24F40/65** (20200101); **G08B3/10** (20130101); **H04W4/029** (20180201);

Field of Classification Search

CPC: G08B (21/24); G08B (3/10); A24F (40/60); A24F (40/65); H04W (4/029)

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Background/Summary

TECHNICAL FIELD

(1) Example embodiments generally relate to non-combustible aerosol provision systems and, in particular, relate to a locator device for use with an non-combustible aerosol provision device.

BACKGROUND

(2) Non-combustible aerosol provision systems (e.g., e-cigarettes/tobacco heating products or other such devices) generally contain an aerosolizable material, such as a reservoir of a source liquid containing a formulation. The formulation typically includes nicotine, or a solid material such as a tobacco-based product, from which an aerosol is generated for inhalation by a user, for example through heat vaporization. However, devices including formulations with other materials, such as cannabinoids (e.g., Tetrahydrocannabinol (THC) and/or Cannabidiol (CBD)), botanicals, medicinals, caffeine, and/or other active ingredients, are also possible. Thus, a non-combustible aerosol provision system will typically include an aerosol generation chamber containing a vaporizer, e.g., a heater, arranged to vaporize a portion of the aerosolizable material to generate an aerosol in the aerosol generation chamber. As a user inhales on a mouthpiece of the device and electrical power is supplied to the heater, air is drawn into the device and into the aerosol generation chamber where the air mixes with the vaporized aerosolizable material and forms a condensation aerosol. There is a flow path between the aerosol generation chamber and an opening in the mouthpiece so the air drawn through the aerosol generation chamber continues along the flow path to an opening in the mouthpiece, carrying some of the condensation aerosol with it, and out through the opening in the mouthpiece for inhalation by the user.

(3) Aerosol provision systems include, for example, vapor products, such as those delivering nicotine that are commonly known as “electronic cigarettes,” “e-cigarettes” or electronic nicotine delivery systems (ENDS), as well as heat-not-burn products including tobacco heating products (THPs). Many of these products take the form of a system including a device and a consumable, and it is the consumable that includes the material from which the substance to be delivered originates. Typically, the device is reusable, and the consumable is single-use (although some consumables are refillable as in the case of so called “open” systems). Therefore, in many cases, the consumable is sold separately from the device, and often in a multipack. Moreover, subsystems and some individual components of devices or consumables may be sourced from specialist manufacturers.

(4) A common trait amongst these types of devices is that they are generally sized to be handheld for usability and transport. However, this size can also be a disadvantage in situations where a user misplaces the device or forgets where the device was last used.

BRIEF SUMMARY OF SOME EXAMPLES

(5) In an example embodiment, an attachable accessory for device location of an aerosol generation device may be provided. The accessory may include a sleeve portion, a power module, a listening

module and an alerting module. The sleeve portion may be configured to receive a portion of the aerosol generation device to operably couple the accessory to the aerosol generation device. The listening module and the alerting module may be powered via the power module. The listening module may be configured to receive an alert signal from a user. The alerting module may be configured to issue an alert responsive to receipt of the alert signal by the listening module.

(6) In another example embodiment, a device locator system may be provided. The system may include an aerosol generation device, an accessory configured to be operably coupled to the aerosol generation device, and a power module. The accessory may include a listening module and an alerting module. The power module may be configured to power the listening module and the alerting module. The listening module may be configured to receive an alert signal from a user. The alerting module may be configured to issue an alert responsive to receipt of the alert signal by the listening module.

(7) It will be appreciated that this Brief Summary is provided merely for purposes of summarizing some example implementations so as to provide a basic understanding of some aspects of the disclosure. Accordingly, it will be appreciated that the above described example implementations are merely examples and should not be construed to narrow the scope or spirit of the disclosure in any way. Other example implementations, aspects and advantages will become apparent from the following detailed description taken in conjunction with the accompanying drawings which illustrate, by way of example, the principles of some described example implementations.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

- (1) Having thus described some example embodiments in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:
- (2) FIG. 1 illustrates a general block diagram of a non-combustible aerosol provision system that may be used in connection with an example embodiment;
- (3) FIG. 2 illustrates a schematic representation of a partially cutaway view of an aerosol provision device that may be used in connection with an example embodiment;
- (4) FIG. 3 is a schematic view of the device of FIG. 2 being operably coupled to an attachable accessory for locating the device using a wireless alert signal in accordance with an example embodiment;
- (5) FIG. 4 illustrates specific form factors that may be employed in accordance with an example embodiment; and
- (6) FIG. 5 illustrates a schematic view of the device of FIG. 2 being operably coupled to an attachable accessory for locating the device using an audible alert signal in accordance with an example embodiment.

DETAILED DESCRIPTION

- (7) Some example embodiments now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all example embodiments are shown. Indeed, the examples described and pictured herein should not be construed as being limiting as to the scope, applicability or configuration of the present disclosure. Rather, these example embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like reference numerals refer to like elements throughout. Furthermore, as used herein, the term “or” is to be interpreted as a logical operator that results in true whenever one or more of its operands are true. As used herein, operable coupling should be understood to relate to direct or indirect connection that, in either case, enables functional interconnection of components that are operably coupled to each other.
- (8) As indicated above, non-combustible aerosol provision systems such as an ENDS device may, due to its size and form factor, become lost or misplaced from time to time. In order to facilitate

location of the device, when lost or misplaced, some example embodiments may provide a locator assembly that can be attached to the device, and which can interact with a locating device (e.g., a smart phone or tablet) through wireless communication. When the locating device is used, the locating device may instruct the locator assembly to initiate an alert to facilitate user location of the locator assembly, and therefore also the device to which the locator assembly is attached.

(9) Given that example embodiments may be employed in connection with location of non-combustible aerosol provision systems such as ENDS devices, a general description of an example device will be provided since some aspects of the locator assembly may be tailored to interface with such devices. In this regard, FIG. 1 illustrates a general block diagram of a non-combustible aerosol provision system that may be used in connection with an example embodiment.

Meanwhile, FIG. 2 illustrates a schematic representation of a partially cutaway view of an ENDS device that may be used in connection with an example embodiment.

(10) Referring first to FIG. 1, a non-combustible aerosol provision system **100** may include a housing **110** inside which a power source **120** and control circuitry **130** may be housed. The housing **110** may further include an aerosol production assembly **140** and an aerosol precursor container **150** inside which an aerosol precursor material (e.g., aerosolizable material) may be stored or contained. The housing **110** may be a single structure, or may be formed from two or more portions that are may be removable with respect to each other. For example, in an open system, the housing **110** may be a single structure with the aerosol precursor container **150** being refillable. However, for a closed system, the housing **110** may include at least one portion inside which the aerosol precursor container **150** is located, and when the aerosol precursor material is exhausted, the portion inside which the aerosol precursor container **150** is located may be removed for replacement with a new or full aerosol precursor container **150**. In some examples with a removable portion inside which the aerosol precursor container **150** is located, the removable portion may be referred to as a cartridge.

(11) The control circuitry **130** may be configured to detect or sense a puff event initiated by a user, and in response to detecting the puff event, the control circuitry **130** may actuate the aerosol production assembly **140** to transform the aerosol precursor material into an aerosol. The control circuitry **130** may therefore include a pressure sensor, a flow sensor, and/or any other suitable devices that can be configured to detect the puff event. A mouthpiece **152** defining an opening **154** in the housing **110** may be associated with the aerosol precursor container **150**, and may be used by the user to initiate the puff event by inhaling at the mouthpiece **152**. Accordingly, in response to the detection of the puff event, the aerosol may be produced by the aerosol production assembly **140** and delivered orally to the user via the mouthpiece **152**.

(12) The aerosol production assembly **140** may be configured to produce the aerosol from the aerosol precursor material using any suitable means. For example, the aerosol production assembly **140** may be embodied as a heat-not-burn device via which, for example, the aerosol is produced by exposing the aerosol precursor material to a heating element (e.g., an induction heater, conduction heater, dielectric heater, microwave heater, radiant heater, arc heater, electrical resistance heater, etc.). In such an example, the aerosol precursor material may be provided in a consumable that may be exposed to the aerosol production assembly **140** such that the heat thereof causes production of the aerosol from the aerosol precursor material. In some cases, the aerosol precursor material may include a substrate and/or a susceptor to facilitate the heating and aerosol release. Alternatively, in the case of a no-heat-no-burn device (e.g., nebulizer), the aerosol production assembly **140** may be embodied as or include a vibratable piezoelectric or piezomagnetic mesh. However, compressed gas, ultrasonic waves, surface acoustic waves, and other technologies may alternatively be employed. The nebulizer may be configured to break up the aerosol precursor material into an aerosol without heating the aerosol precursor material. In other words, heat generation may or may not be involved in the operation of the aerosol production assembly **140**. Moreover, in some cases, the aerosol production assembly **140** may include a combination of elements, which can include

both a heating element and an additional element, such as a vibrating aerosol production component (e.g., a vibratable piezoceramic and/or other piezoelectric or piezomagnetic material) that cooperate to produce aerosol from an aerosol precursor material. Such combinations may be referred to as hybrid products.

(13) The aerosol precursor material may be a solid, semi-solid, or liquid material. As such, the aerosol precursor container **150** may be configured to retain the aerosol precursor material in whatever form such material may take. In some cases, the aerosol precursor container **150** may be a reservoir configured to store liquid that is operably coupled to the aerosol production assembly **140** (e.g., directly or indirectly) for the generation of the aerosol as described above. In some examples, the aerosol precursor material may be provided in a substrate (e.g., coated or absorbed on/in the substrate) such that the aerosol precursor material may be integrated in, stored in, or deposited on the substrate prior to being used for generation of the aerosol.

(14) The power source **120** may be a replaceable or rechargeable battery. Rechargeable batteries may be useful to avoid or limit production of waste materials, and to facilitate ease of operation. To facilitate recharging of the power source **120**, a charge interface **122** may be provided. The charge interface **122** may include a USB (Universal Serial Bus) port or other charge port into which a charger cord or other charging device may be plugged or inserted. The charge interface **122** may therefore form a penetration or opening in the housing **110**.

(15) FIG. 2 is a cross-sectional view through one example non-combustible aerosol provision device **200** that may be implemented in connection with an example embodiment. The non-combustible aerosol provision device **200** is one more detailed example of the non-combustible aerosol provision system **100** of FIG. 1. The non-combustible aerosol provision device **200** of FIG. 2 is a two-part device (i.e., a closed system), which includes a control unit **210** and a cartridge **220**. The cartridge **220** may be referred to as a consumable part (or replaceable/disposable part) and the control unit **210** may be referred to as a reusable part.

(16) In normal use the control unit **210** and the cartridge **220** may be releasably coupled together at a coupling interface **230**. When the cartridge **220** is exhausted or the user simply wishes to switch to a different cartridge **220** (e.g., for a different flavor), the cartridge **220** may be removed from the control unit **210** and a replacement (i.e., a different or new instance of the cartridge **220**) may be attached to the control unit **210** in place of the original cartridge **220**. The coupling interface **230** may provide a structural, electrical and/or air path connection between the cartridge **220** and the control unit **210**, and may be established in accordance with conventional techniques, which may include a screw thread, latch mechanism, or bayonet fixing with appropriately arranged electrical contacts and openings for establishing the electrical connection and air path between the cartridge **220** and the control unit **210** as appropriate. The specific manner by which the cartridge **220** mechanically mounts to the control unit **210** is not significant to the principles described herein, but for the sake of a concrete example is assumed here to comprise a latching mechanism, for example with a portion of the cartridge **220** being received in a corresponding receptacle in the control unit **210** with cooperating latch engaging elements. It will also be appreciated the coupling interface **230** in some implementations may not support an electrical connection between the respective parts. For example, in some implementations a vaporiser may be provided in the control unit **210** rather than in the cartridge **220**.

(17) The cartridge **220** may include a consumable housing **222** (e.g., as an example of the aerosol precursor container **150** of FIG. 1). The consumable housing **222** may be formed of a plastic, composite or metallic material. The consumable housing **222** may support other components of the cartridge **220** and provide support for a portion of the mechanical coupling interface **230** with the control unit **210**. The consumable housing **222** in this example is generally circularly symmetric about a longitudinal axis along which the cartridge **220** couples to the control unit **210**. The consumable housing **222** of this example may have a length of about 4 cm and a diameter of around 1.5 cm. However, it will be appreciated the specific geometry, and more generally the overall

shapes and materials used, may be different in different implementations.

(18) A reservoir **224** may be provided within the consumable housing **222** to contain liquid aerosolisable material (e.g., the aerosol precursor material of FIG. 1). The aerosolisable material may be referred to as e-liquid in some examples. The liquid reservoir **224** in this example has an annular shape, though it will be appreciated that other shapes are within the scope of the disclosure, with an outer wall defined by the consumable housing **222** and an inner wall that defines an air path **226** through the cartridge **220**. The reservoir **224** is closed at each end with end walls to contain the e-liquid. The reservoir **224** may be formed in accordance with conventional techniques and may, for example, be formed of a plastic material integrally molded with the consumable housing **222**. The opening of the air path **226** at the end of the cartridge **220** provides a mouthpiece outlet **228** for the non-combustible aerosol provision system through which a user inhales aerosol generated by the non-combustible aerosol provision device **200** during use.

(19) The cartridge **220** of this example may further include a wick **250** and a heater element **260** (e.g., a vaporiser) located proximate to an end of the reservoir **224** opposite to the mouthpiece outlet **228**. In this example the wick **250** extends transversely across the air path **226** with ends thereof extending into the reservoir **224** of e-liquid through openings in the inner wall of the reservoir **224**. The openings in the inner wall of the reservoir **224** may be sized to broadly match the dimensions of the wick **250** to provide a reasonable seal against leakage from the reservoir **224** into the air path **226** without unduly compressing the wick **250**, which may be detrimental to fluid transfer performance.

(20) The wick **250** and heater element **260** may be arranged in the air path **226** of the cartridge **220** such that a region of the air path **226** around the wick **250** and heater element **260** in effect defines a vaporisation region for the cartridge **220**. E-liquid in the reservoir **224** infiltrates the wick **250** through the ends of the wick **250** that extend into the reservoir **224** and is drawn along the wick **250** by surface tension/capillary action (i.e. wicking). The heater element **260** in this example may be embodied as an electrically resistive wire coiled around the wick **250**. In this example, the wick **250** may be a glass fibre bundle, but other configurations are also possible. In use, electrical power may be supplied to the heater element **260** to vaporise an amount of e-liquid (e.g., aerosolisable material) drawn to the vicinity of the heater element **260** by the wick **250**. Vaporised e-liquid may then become entrained in air drawn along the air path **226** from the vaporisation region to form a condensation aerosol that exits the system through the mouthpiece outlet **228** for user inhalation. Thus electrical power can be applied to the heater element **260** to selectively generate aerosol from the e-liquid in the cartridge **220**. When the device is in use and generating aerosol, the amount of power supplied to the heater element **260** may be varied, for example through pulse width and/or frequency modulation techniques, to control the temperature and/or rate of aerosol generation as desired.

(21) The control unit **210** may include an outer housing **212** (e.g., as a portion of the housing **110** of FIG. 1) with an opening that defines an air inlet **214** for the non-combustible aerosol provision device **200**. The non-combustible aerosol provision device **200** may also include, within the outer housing **212**, a battery **270** for providing operating power for the non-combustible aerosol provision device **200**. The battery **270** may be operably coupled to control circuitry **280** configured for controlling and monitoring the operation of the non-combustible aerosol provision device **200**. The battery **270** may be an example of the power source **120**, and the control circuitry **280** may be an example of the control circuitry **130** of FIG. 1.

(22) The control circuitry **280** may be operably coupled to an inhalation sensor **282** (e.g., puff detector), which in this example comprises a pressure sensor located in a pressure sensor chamber **284**. The control circuitry **280** may also be operably coupled to a visual display **286** (which may be optional). The visual display **286** may include one or more lights configured to indicate various status conditions of the non-combustible aerosol provision device **200** based on light color, flash sequences, or other indications. Alternatively or additionally, the visual display **286** may be

configured to display characters, images and/or the like via a liquid crystal display (LCD) screen, one or more light emitting diodes (LEDs) or other display options. Thus, the visual display **286** may be provided to give a user a visual indication of various characteristics associated with the non-combustible aerosol provision device **200**. For example, the visual display **286** may provide information indicative of current power and/or temperature setting information, remaining battery power, and so forth. As an alternative (or in addition) to the visual display **286**, some example embodiments may include other means for providing a user with information relating to operating characteristics of the non-combustible aerosol provision device **200** such as, for example, using audio signalling or haptic feedback.

(23) The control circuitry **280** may be configured to monitor the output from the inhalation sensor **282** to determine when a user is inhaling through the mouthpiece opening **228** of the cartridge **220** so that power can be automatically supplied to the heating element **260** to generate aerosol in response to user inhalation. In other implementations, as an alternative to automatic operation of the heating element **260**, a button **288** may be provided instead of the inhalation sensor **282**, and power may be supplied to the heating element **260** in response to a user manually activating the button **288** to trigger aerosol generation. Thus, the button **288** may also be entirely optional and omitted in some cases.

(24) The outer housing **212** may be formed, for example, from a plastic or metallic material and may be shaped to have any desirable profile. In some examples, the outer housing **212** may be substantially cylindrical and therefore have a circular cross-section generally conforming to the shape and size of the cartridge **220** so as to provide a smooth transition between the two parts at the coupling interface **230**. In some examples, the control unit **210** may have a length of around 8 cm so the overall length of the non-combustible aerosol provision device **200** when the cartridge **220** and control unit **210** are operably coupled together is around 12 cm. However, and as already noted, it will be appreciated that the overall shape and scale of components may be changed in different example embodiments without altering the principles described herein.

(25) The air inlet **216** connects to an air path **216** through the control unit **210**. The air path **216** of the control unit **210** in turn connects to the air path **226** of the cartridge **220** across the coupling interface **230** when the control unit **210** and cartridge **220** are operably coupled together. The pressure sensor chamber **284** containing the pressure sensor **282** may be in fluid communication with the air path **216** in the control unit **210** (i.e. the pressure sensor chamber **284** branches off from the air path **214** in the control unit **210**). Thus, when a user inhales on the mouthpiece opening **228**, there is a drop in pressure in the pressure sensor chamber **284** that may be detected by the pressure sensor **282** and also air is drawn in through the air inlet **214**, along the air path **216** of the control unit **210**, across the coupling interface **230**, through the aerosol generation region in the vicinity of the heating element **260** (where an aerosol generated from the aerosolisable material becomes entrained in the air flow when the heating element **260** is active), along the air path **226** of the cartridge **220**, and out through the mouthpiece opening **228** for user inhalation.

(26) The battery **270** in this example is rechargeable and may be recharged via charging connector **290**. In this regard, the battery **270** may be recharged through an opening in the control unit housing **212** at which the charging connector **290** is formed, and to which a charging plug or other charging device may be operably coupled. The charging connector **290** may take any suitable configuration including, for example, a USB connector, other standard power connectors, or even proprietary charging connections.

(27) The control circuitry **280** may be configured or programmed to control the operation of the non-combustible aerosol provision device **200** to provide various functions thereof. The control circuitry **280** may be considered to logically comprise various sub-units or circuitry elements associated with different aspects of the operation of the non-combustible aerosol provision device **200** in accordance with the principles described herein and other conventional operating aspects of non-combustible aerosol provision devices **200**, such as display driving circuitry and user input

detection. It will be appreciated the functionality of the control circuitry **280** can be provided in various different ways such as, for example, using one or more suitably programmed programmable computer(s) and/or one or more suitably configured application-specific integrated circuit(s)/circuitry/chip(s)/chipset(s) configured to provide the desired functionality.

(28) In some cases, the non-combustible aerosol provision device **200** may have three basic operating states. However, additional or different operating states are also possible. The three basic operating states may include an “off” state, an “on” state, and a “standby” state. In the off state, the non-combustible aerosol provision device **200** may be unable to generate aerosol (i.e. the control circuitry **280** may prevent supplying of power to the heating element **260** in the off state). The non-combustible aerosol provision device **200** may, for example, be placed in the off state between use sessions, for example, when the non-combustible aerosol provision device **200** might be set aside or placed in a user's pocket or bag. In the on (or active) state, the non-combustible aerosol provision device **200** may be enabled to actively generate aerosol (e.g., the control circuitry **280** may provide (or enable provision of) power to the heating element **260**). The non-combustible aerosol provision device **200** will thus typically be in the on state when a user is in the process of inhaling aerosol from the non-combustible aerosol provision device **200**. In the standby state the non-combustible aerosol provision system may be ready to generate aerosol (e.g., ready to apply power to the heater element **260** of the illustrated embodiment) in response to user activation, but may not currently be doing so. The non-combustible aerosol provision device **200** will typically be in the standby state when a user initially exits the off state to begin a session of use (e.g., when a user initially turns on the non-combustible aerosol provision device **200**), or between uses during an ongoing session of use (e.g., between puffs when the user is using the non-combustible aerosol provision device **200**). It is more common for examples of the non-combustible aerosol provision device **200** using liquid aerosolisable material to revert to the standby mode between puffs, whereas non-combustible aerosol provision devices **200** using solid aerosolisable material may more often remain on between puffs to seek to maintain the aerosolisable material at a desired temperature during a session of use comprising a series of puffs.

(29) To generate an aerosol in the non-combustible aerosol provision device **200**, electrical power from the battery **270** is supplied to the heater element **260** under control of the control circuitry **280**. When the non-combustible aerosol provision device **200** is on, i.e. actively generating an aerosol, power may be supplied to the heater element **260** in a pulsed fashion, for example, using a pulse width modulation (PWM) scheme to control the level of power being delivered. Thus, the power supplied to the heater element **260** during a period of aerosol generation may comprise an alternating sequence of on periods during which power is connected to the electric heater and off periods during which power is not connected to the heater element **260**. The cycle period for the pulse width modulation (i.e. the duration of a neighbouring pair of an off and an on period) is in this example 0.020 s (20 ms) (i.e. the pulse width modulation frequency is 50 hertz). The proportion of each cycle period during which power is being supplied to the heater (i.e. the length of the on period) as a fraction of the cycle period is the so-called duty cycle for the pulse width modulation. In accordance with certain embodiments of the disclosure, the control circuitry of the non-combustible aerosol provision system may be configured to adjust the duty cycle for the pulse width modulation to vary the power supplied to the heater, for example to achieve a target level of average power or to achieve a target temperature.

(30) As noted above, when the non-combustible aerosol provision device **200** is in the off state, it is common for the user to store the non-combustible aerosol provision device **200** in a pocket or bag, or otherwise set the non-combustible aerosol provision device **200** aside. Invariably, some users will forget where they last put or saw the non-combustible aerosol provision device **200**, and a search for the device may prove necessary before it can be located. These situations can be frustrating, particularly when the resulting search is extended or unsuccessful. Accordingly, it may be desirable to provide a way by which to ensure that the non-combustible aerosol provision device

200 can be located quickly when its location is not immediately apparent. Some example embodiments may address this issue by providing an attachable accessory **300** that makes the non-combustible aerosol provision device **200**, when the attachable accessory **300** is attached thereto, trackable or discoverable. An example of the attachable accessory **300** is shown and will be described in reference to FIG. 3.

(31) In an example embodiment, the attachable accessory **300** may be configured specifically for connection to the non-combustible aerosol provision device **200** in order to retain a sleek and appealing appearance. However, the attachable accessory **300** may further be configured to provide additional functionality that enables location of the non-combustible aerosol provision device **200**. As such, when connected together, the combination of the non-combustible aerosol provision device **200** and the attachable accessory **300** may form a device locator system for locating the non-combustible aerosol provision device **200**.

(32) In some example embodiments, the attachable accessory **300** may be configured to communicate wirelessly with a wireless communication device **310** such as a cell phone, laptop or tablet. However, a dedicated paging device may also be employed in some examples. In some embodiments in which a dedicated paging device is employed, the dedicated paging device may, for example, have the form factor of a key fob. Regardless of the specific form of the wireless communication device **310**, the user may be able to use the wireless communication device **310** to trigger the attachable accessory **300** to initiate an alert that may enable the user to locate the attachable accessory **300**, and therefore also the non-combustible aerosol provision device **200**. As such, for example, the alert may often be audible, although the alert may also or alternatively include visual or haptic outputs as well. For example, visual alerts could include an indication of a location of or to the device being provided (e.g., on the display of the cell phone, laptop, tablet, etc.) For example, a relative location indication (e.g., directional pointer) may be provided and the indication may get stronger as the user gets closer to the location of the device (and weaker as the user gets farther from the device). The same strategy regarding strength of indication could also be employed for audible (e.g., louder and/or more frequent audible sound as the user gets closer to the location of the device and softer and/or less frequent audible sound as the user gets farther from the device) and haptic feedback (e.g., stronger and/or more frequent haptic feedback as the user gets closer to the location of the device and softer and/or less frequent haptic feedback as the user gets farther from the device). Relative location could also or alternatively be indicated visually in manners ranging from distance between two points—e.g., one representative of aerosol device and one of mobile device/user; a color representative of relative distance (e.g., red for relatively longer distance, green for close distance, etc.); intensity of light (e.g., brighter for closer proximity), etc.

(33) The attachable accessory **300** may include a housing **320** that is configured to engage, mate with or otherwise interface with the non-combustible aerosol provision device **200** in a way that securely attaches the respective devices together. In this regard, in some cases, the attachable accessory **300** may be constructed to include a sleeve portion **322** formed in the housing **320**. The sleeve portion **322** may have an internal diameter and/or shape that is configured to substantially match an external shape, such as a diameter or other cross-sectional profile, of the control unit **210** (or another portion of the non-combustible aerosol provision device **200**). However, the sleeve portion **322** could engage the non-combustible aerosol provision device **200** in other ways, and take numerous forms. For example, a distal end of the control unit **210** (relative to the cartridge **220**) may be configured to slide into and be received inside the sleeve portion **322** as shown by arrow **324** in FIG. 3. A cross-sectional profile (e.g., diameter) of the sleeve portion **322** may taper slightly as it proceeds inwardly into the housing **320** such that the control unit **210** and the sleeve portion **322** may stay in contact with each other by friction responsive to insertion of the control unit **210** into the sleeve portion **322**. However, in alternative embodiments, other fixing methods (including latching mechanisms and/or the like) may be employed.

(34) In this regard, for example, the housing **320** may include portions that are attached via a hinge

that can be opened to allow the non-combustible aerosol provision device **200** to be inserted into the housing **320**, and then closed to retain the non-combustible aerosol provision device **200** in the housing **320**. The hinge may be a separate component, or may be a living hinge (i.e., formed from the same material forming the portions of the housing **320**, which are joined). Another alternative example may include a case with flexible sides or portions. The sides or portions may be biased toward a closed (engagement) position, such that the sides or portions can be manipulated away from the closed (engagement) position to an open position in order to facilitate insertion of and/or engagement with the non-combustible aerosol provision device **200**. When the non-combustible aerosol provision device **200** has been positioned in the housing **320**, the force used to manipulate the sides or portions to the open position may be removed, and the sides or portions may release and return to the closed (engagement) position to retain the non-combustible aerosol provision device **200** in the housing **320**.

(35) The housing **320** and sleeve portion **322** may be configured to leave any desirable portions of the control unit **210** (and all of the cartridge **220**) exposed. Thus, for example, the sleeve portion **322** may not extend along the control unit **210** as far as the coupling interface **230**. Moreover, to the extent the visual display **286** and/or button **288** are included on the control unit **210**, the sleeve portion **322** and the housing **320** may be configured to leave both the visual display **286** and/or button **288** exposed. Accordingly, the attachable accessory **300** may be attached to the control unit **210** without negatively inhibiting the functionality of the non-combustible aerosol provision device **200** in any way.

(36) As shown in FIG. 3, the attachable accessory **300** may include a number of modules that may be configured to perform respective functions of the attachable accessory **300**. Although it may be possible to power the attachable accessory **300** from the battery **270** of the non-combustible aerosol provision device **200** (or the power source **120** of non-combustible aerosol provision system **100** of FIG. 1), doing so may drain the battery **270** faster, and cause users to be dissuaded from using the attachable accessory **300**. Accordingly, it may instead be desirable to power the attachable accessory **300** via its own separate power source. Thus, the attachable accessory **300** may include a power module **330**.

(37) In some examples, the power module **330** may be a rechargeable or replaceable battery. Moreover, in some cases, the power module **330** may be a Lithium ion battery or other battery that can provide significant power in a relatively small form factor. By providing the power module **330** as a separate power source relative to the battery **270** of the non-combustible aerosol provision device **200**, not only may the functioning of the non-combustible aerosol provision device **200** not be negatively impacted by operation and current usage by the attachable accessory **300**, but the non-combustible aerosol provision device **200** may be located even when powered off, or when the battery **270** is dead.

(38) In some cases, the power module **330** may not only be separate from the battery **270** (i.e., two respective separate battery packs or cells), but the power module **330** could also be entirely independent of the battery **270**. In other words, there may be no connection between the battery **270** of the non-combustible aerosol provision device **200** and the power module **330** of the attachable accessory **300**. Thus, the battery **270** of the non-combustible aerosol provision device **200** may only provide power to the non-combustible aerosol provision device **200** and the power module **330** of the attachable accessory **300** may only provide power to the attachable accessory **300**. In such a case, the attachable accessory **300** may be constructed to avoid interference with accessibility of the charging connector **290** of the non-combustible aerosol provision device **200** to enable recharging of the battery **270** even when the control unit **210** is mated with the attachable accessory **300**. The power module **330** may therefore include a charge port **332** that enables the power module **330** to be charged whether or not the power module **330** is operably coupled to the control unit **210**, and both the battery **270** and the power module **330** could be separately (and possibly even simultaneously) charged via the charging connector **290** and the charge port **332**, respectively.

(39) In other examples, the battery **270** and the power module **330** may not be independent of each other. To the contrary, for example, the power module **330** may power the attachable accessory **300**, but also be capable of powering the non-combustible aerosol provision device **200** either directly (e.g., by connection to the control circuitry **280**) or indirectly (e.g., by charging the battery **270**). In this regard, for direct powering, the power module **330** may provide power to the control circuitry **280** when the battery **270** is either dead or below a threshold level of charge. For indirect powering, the power module **330** may be configured to interface with the battery **270** to charge the battery **270**. For example, the power module **330** may include a charging interface **334** configured to mate with the charging connector **290** of the control unit **210** when the control unit **210** is inserted into the sleeve portion **322**. When the charging interface **334** is mated with the charging connector **290**, the power module **330** may be operably coupled to the battery **270** to enable the battery **270** to be charged. Thus, for example, the power module **330** may be configured to supply a higher voltage than the battery **270** so that, when operably coupled to each other, the battery **270** may be charged.

(40) In such an example, the control circuitry **280** operation may be unimpeded in that the control circuitry **280** may still only supply power to the heater element **260** responsive to detecting the puff event, and the power may be still supplied from the battery **270**. However, the battery **270** could be charged either simultaneously or at other (i.e., non-operational) times for the heater element **260**. Thus, in some cases, charging from the power module **330** to the battery **270** may only be possible when the non-combustible aerosol provision device **200** is not operating. In some cases, the power module **330** may only charge the battery **270** when the non-combustible aerosol provision device **200** is in the off state. In other cases, charging from the power module **330** to the battery **270** may only be possible when the non-combustible aerosol provision device **200** is in the off state or the standby state. However, it is also possible, as noted above, for charging during the on state in some cases. In some cases, the power module **330** may be configured to receive information indicative of the state of the non-combustible aerosol provision device **200**, and control charging of the battery **270** (or provision of power directly to the control circuitry **280**) based on the information received. Thus, for example, charging could be stopped when the non-combustible aerosol provision device **200** is active and/or in the standby status.

(41) As shown in FIG. 3, the attachable accessory **300** may further include a listening module **340** and an alerting module **350**. The listening module **340** may take multiple forms in accordance with different optional embodiments. In this regard, for example, in the example of FIG. 3, the listening module **340** may include or otherwise be embodied as a wireless receiver **342** configured to communicate with the wireless communications device **310**. Although the listening module **340** may, in some cases, include a transceiver to permit two-way communication with the wireless communications device **310**, such two-way communication is not necessary in all embodiments. Thus, in some embodiments, only one-way communication (i.e., from the wireless communications device **310** to the listening module **340**) may be supported in order to reduce complexity and cost of the attachable accessory **300**, and increase the battery life of the power module **330**.

(42) The wireless receiver **342** (or transceiver) of the listening module **340** may include one or more antennas, and corresponding radio circuitry configured to enable the listening module **340** to receive and process wireless signals (e.g., wireless alert signal **370**). As such, the wireless receiver **342** of the listening module **340** may be ready to receive wireless communication signals at any time while receiving power from the power module **330**. The readiness to receive such signals may be continuous, or may be periodic (e.g., with periodic cyclic listening periods being defined) to reduce battery power consumption. In some cases, the wireless communication between the listening module **340** and the wireless communications device **310** may be conducted via a short range wireless communication protocol (e.g., Bluetooth, Zigbee, WiFi, etc.).

(43) The alerting module **350** may be configured to generate a sound, light, and/or haptic output in response to triggering from the listening module **340**, which may (as noted above) further include information indicative of proximity and/or relative location based, for example, on two-way

communication and signal strength measurement with or without directional discrimination capability. In this regard, responsive to receiving communications from the wireless communications device **310** (either at all, or specifically requesting an alert be issued), the alerting module **350** may issue the alert. In an example embodiment, the wireless communications device **310** may be configured to execute an application or function that sends a signal to the wireless receiver **342** of the listening module **340**. For example, the user may launch an application at the wireless communications device **310** for locating the non-combustible aerosol provision device **200** (e.g., a “find my device” app). When launched, the application may include one or more simple commands for controlling operation of the components of the device locator system. For example, the application may include a locate device command and a stop locating command. In response to user selection of the locate device command, the wireless communications device **310** may send a trigger signal or a continuous alerting signal to the wireless receiver **342** of the listening module **340**. In response to the trigger signal or continuous alerting signal, the wireless receiver **342** of the listening module **340** may signal the alerting module **350** to begin generation of the alert.

(44) In some cases, the alerting module **350** may be configured to continue to generate the alert for as long as the wireless receiver **342** of the listening module **340** is in receipt of the continuous alerting signal. The continuous alerting signal may be communicated from the wireless communications device **310** until the device is found and the user selects an icon or button on the wireless communications device **310** to stop sending the continuous alerting signal **310**. However, if the trigger signal is instead employed, the alerting module **350** may be configured to continue to generate the alert after receipt of the trigger signal until another signal (either a stop signal, or a second instance of the trigger signal) is received to indicate that the device has been found or located. As an alternative, the alerting module **350** may be configured to provide the alert until a predetermined period of time (e.g., 10 seconds, 20 seconds, 30 seconds, etc.) has passed. After the predetermined period of time has passed, either the device will be found, or another user selection of the locate device command at the wireless communications device **310** may be issued if continued searching (and alerting) is desired.

(45) FIG. 4 illustrates a specific example of form factor and device options that may be employed to define a device locator system **400** in accordance with an example embodiment. In this regard, FIG. 4 illustrates an aerosol generation device **410** (e.g., an example of either the non-combustible aerosol provision system **100** of FIG. 1 or the non-combustible aerosol provision device **200** of FIGS. 2 and 3). The aerosol generation device **410** includes a control unit **412** and cartridge **414** that are similar to the components of the same name discussed above in reference to FIGS. 2 and 3. The control unit **412** of this example does not have a circular cross section, but instead has a substantially oval cross section, which demonstrates the variability in shape and size that may exist for control units. In any case, attachable accessory **420** is configured to interface with the control unit **412** similar to the manner described above in reference to FIG. 3. Moreover, the attachable accessory **420** includes a charge port **430** and a speaker **440**. The charge port **430** may be used for charging the battery of the attachable accessory **420**, and the speaker **440** may be an example of an alerting device used to issue the alert (i.e., a whistle, beep, tone, or other audible alerting sound).

(46) The attachable accessory **420** may also include a wireless receiver that triggers actuation of the speaker **440** responsive to receipt of an alert signal (e.g., a trigger signal or continuous alerting signal described above) from a smart phone **450**. The smart phone **450** may be an example of a wireless communication device **310**, and a display **452** thereof may provide commands (e.g., start command **454** and stop command **456**) for selection by the user.

(47) The user of the smart phone **450** may select an icon to launch a device locator application, and the interface screen shown on the display **452** of FIG. 4 (or something similar) may be presented. The user may then select the start command **454**, which may cause the smart phone **450** to issue the alert signal to the attachable accessory **420**. The wireless receiver on the attachable accessory **420** may then trigger the speaker **440** to generate the alert. If within earshot, the user may then follow

the sound of the alert until the attachable accessory **420** (and therefore also the aerosol generation device **410** to which the attachable accessory **420** is attached) is located. Once the attachable accessory **420** is located, the stop command **456** may be selected at the smart phone **450** in order to stop generation of the alert. As noted above, the alert may time out in some cases if the attachable accessory **420** is not found within a predetermined period of time.

(48) As noted above, the listening module **340** can also take other forms. For example, as shown in FIG. 5, a different version of listening module **340'** is provided. Other than the form and function of the listening module **340'** of this example, the attachable accessory **300** may be substantially similar or identical to that described above. Meanwhile, the listening module **340'** may not receive wireless signals, but may instead receive audible signals. As such, the listening module **340'** may include a microphone or other audio sensor device. Thus, rather than “listening” for a wireless communication signal as an alert signal (as was the case for the examples of FIGS. 3 and 4), the listening module **340'** of FIG. 5 may be configured to “listen” for an audible form of the alert signal (i.e. audible alert signal **500**).

(49) The audible form of the alert signal may be a key word, key sound, or even a tone that could be played by the wireless communication device **310**. In this regard, for example, the application mentioned above may cause the wireless communication device **310** to issue the alert signal in the form of an audible key word, key sound or tone that is received at the listening module **340'** to trigger the alerting module to issue the alert as described above. Since the alert signal is audible, it should also be appreciated that the alert signal could directly be issued by the user. For example, the user may speak the key word or key sound to trigger the alert from the attachable accessory **300** as long as the attachable accessory **300** is within range of the user and can receive the audibly generated alert signal. In some cases, the user may pre-record the key word or key sound to program the listening module **340'** to recognize the key word or key sound.

(50) Accordingly, as can be appreciated from the examples above, an attachable accessory for device location of an aerosol generation device may be provided in accordance with an example embodiment. The accessory may include a sleeve portion, a listening module and an alerting module. The sleeve portion may be configured to engage a portion of the aerosol generation device to operably couple the accessory to the aerosol generation device. The listening module may be configured to receive an alert signal from a user. The alerting module may be configured to issue an alert responsive to receipt of the alert signal by the listening module.

(51) The attachable accessory may include a number of modifications, augmentations, or optional additions, some of which are described herein. The modifications, augmentations or optional additions listed below may be added in any desirable combination. Within this context, the attachable accessory described above may be considered a first embodiment, and other embodiments may be defined by each respective combination of modifications, augmentations or optional additions. For example, a second embodiment may be defined in which the alerting module may include a speaker configured to issue the alert as an audible sound (e.g., whistle, tone or pulse). In an example embodiment, a third embodiment may be defined in which the alerting module may be configured to issue the alert for a predetermined period of time. The third embodiment may be combined with any or all of embodiments one and two. In some cases, a fourth embodiment may be defined in which the alerting module may be configured to issue the alert until an instruction to stop issuing the alert is received. The fourth embodiment may be combined with any or all of embodiments one to three. In an example embodiment, a fifth embodiment may be defined in which the sleeve portion may be formed in a housing of the accessory, the accessory may further include a power module, and the housing may further include a charge port via which the power module is rechargeable from an external power source. The fifth embodiment may be combined with any or all of embodiments one to four. In some cases, a sixth embodiment may be defined in which the power module may be configured to provide power (directly or indirectly) to a battery of the aerosol generation device. The sixth embodiment may be

combined with any or all of embodiments one to five. In an example embodiment, a seventh embodiment may be defined in which the listening module may be configured to receive the alert signal in the form of an audible key word or key sound generated (e.g., uttered or spoken) by the user. The seventh embodiment may be combined with any or all of embodiments one to six. In some cases, an eighth embodiment may be defined in which the audible key word or key sound may be pre-programmed into the listening module by the user. The eighth embodiment may be combined with any or all of embodiments one to seventh. In an example embodiment, a ninth embodiment may be defined in which the listening module may include a wireless receiver configured to communicate wirelessly with a wireless communication device to receive the alert signal as a wireless alert signal sent by the wireless communication device, responsive to a request from the user. The ninth embodiment may be combined with any or all of embodiments one to eight. In some cases, a tenth embodiment may be defined in which the wireless signal may be sent responsive to activation of the wireless communication device via an application executed by the user at the wireless communication device. The tenth embodiment may be combined with any or all of embodiments one to nine. In an example embodiment, an eleventh embodiment may be defined in which the application may include selectable icons for initiating sending of the alert signal and stopping the sending of the alert signal. The eleventh embodiment may be combined with any or all of embodiments one to ten.

(52) Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although the foregoing descriptions and the associated drawings describe exemplary embodiments in the context of certain exemplary combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated as may be set forth in some of the appended claims. In cases where advantages, benefits or solutions to problems are described herein, it should be appreciated that such advantages, benefits and/or solutions may be applicable to some example embodiments, but not necessarily all example embodiments. Thus, any advantages, benefits or solutions described herein should not be thought of as being critical, required or essential to all embodiments or to that which is claimed herein. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. An attachable accessory for device location of an aerosol generation device that includes a power source comprising a battery, control circuitry, and an aerosol production assembly that is supplied power from the power source to transform an aerosol precursor material into an aerosol, the accessory comprising: a sleeve portion configured to engage a portion of the aerosol generation device to operably couple the accessory to the aerosol generation device that is operable to transform the aerosol precursor material to the aerosol whether or not the accessory is operably coupled to the aerosol generation device; a listening module; an alerting module; a power module configured to power the listening module and the alerting module, wherein the power module is operatively connected to the control circuitry of the aerosol generation device when the sleeve portion engages the aerosol generation device and the control circuitry is configured to supply power to the battery of the aerosol generating device from the power module of the accessory when the battery of the aerosol generating device is below a threshold level of charge, wherein the

listening module is configured to receive an alert signal from a user, and wherein the alerting module is configured to issue an alert responsive to receipt of the alert signal by the listening module.

2. The accessory of claim 1, wherein the alerting module comprises a speaker configured to issue the alert as an audible sound.

3. The accessory of claim 1, wherein the alerting module is configured to issue the alert for a predetermined period of time.

4. The accessory of claim 1, wherein the alerting module is configured to issue the alert until an instruction to stop issuing the alert is received.

5. The accessory of claim 1, wherein the sleeve portion is formed in a housing of the accessory, and wherein the housing further comprises a charge port via which the power module is rechargeable from an external power source.

6. The accessory of claim 1, wherein the listening module is configured to receive the alert signal in the form of an audible key word or key sound.

7. The accessory of claim 6, wherein the audible key word or key sound is pre-programmed into the listening module by the user.

8. The accessory of claim 1, wherein the listening module is configured to receive the alert signal in the form of an audible tone played by a locating device, responsive to a request from the user.

9. The accessory of claim 8, wherein the audible tone is played responsive to activation of the locating device via an application executed by the user at the locating device.

10. The accessory of claim 9, wherein the application includes selectable icons for initiating the audible tone and stopping the audible tone.

11. The accessory of claim 1, wherein the listening module is configured to receive an audible alert signal from the user, and wherein the alerting module is configured to issue the alert responsive to receipt of the audible alert signal by the listening module.

12. A device locator system comprising: an aerosol generation device that includes a power source comprising a battery, control circuitry, and an aerosol production assembly that is supplied power from the power source to transform an aerosol precursor material into an aerosol; and an accessory configured to be operably coupled to the aerosol generation device that is operable to transform the aerosol precursor material to the aerosol whether or not the accessory is operably coupled to the aerosol generation device, the accessory comprising a listening module, an alerting module, and a power module configured to power the listening module and the alerting module, wherein the power module is operatively connected to the control circuitry of the aerosol generation device when a sleeve portion engages the aerosol generation device and the control circuitry is configured to supply power to the battery of the aerosol generating device from the power module of the accessory when the battery of the aerosol generating device is below a threshold level of charge, wherein the listening module is configured to receive an alert signal from a user, and wherein the alerting module is configured to issue an alert responsive to receipt of the alert signal by the listening module.

13. The system of claim 12, wherein the accessory comprises the sleeve portion configured to receive a portion of the aerosol generation device to operably couple the accessory to the aerosol generation device, and wherein the alerting module comprises a speaker configured to issue the alert as an audible sound.

14. The system of claim 13, wherein the alerting module is configured to issue the alert for a predetermined period of time, or wherein the alerting module is configured to issue the alert until an instruction to stop issuing the alert is received.

15. The system of claim 12, wherein the sleeve portion is formed in a housing of the accessory, and wherein the housing further comprises a charge port via which the power module is rechargeable from an external power source.

16. The system of claim 15, wherein the power source of the aerosol generation device includes the

battery, and a charging connector from which the battery is rechargeable, and wherein the power module includes a charging interface configured to mate with the charging connector, and from which the power module is configured to provide power to the battery of the aerosol generation device.

17. The system of claim 12, wherein the listening module is configured to receive the alert signal in the form of an audible key word or key sound.

18. The system of claim 12, wherein the listening module is configured to receive the alert signal in the form of an audible tone played by a locating device, responsive to a request from the user.

19. The system of claim 18, wherein the audible tone is played responsive to activation of the locating device via an application executed by the user at the locating device.

20. The system of claim 19, wherein the application includes selectable icons for initiating the audible tone and stopping the audible tone.

21. The system of claim 12, wherein the listening module is configured to receive an audible alert signal from the user, and wherein the alerting module is configured to issue the alert responsive to receipt of the audible alert signal by the listening module.

22. An attachable accessory for device location of an aerosol generation device that includes a power source and an aerosol production assembly that is supplied power from the power source to transform an aerosol precursor material into an aerosol, the accessory comprising: a sleeve portion configured to engage a portion of the aerosol generation device to operably couple the accessory to the aerosol generation device that is operable to transform the aerosol precursor material to the aerosol whether or not the accessory is operably coupled to the aerosol generation device; a listening module; an alerting module; and a power module configured to power the listening module and the alerting module, wherein the sleeve portion is formed in a housing of the accessory, and wherein the housing further comprises a charge port via which the power module is rechargeable from an external power source; wherein the power source of the aerosol generation device includes a battery, and a charging connector from which the battery is rechargeable, and wherein the power module includes a charging interface configured to mate with the charging connector, and from which the power module is configured to provide power to the battery of the aerosol generation device, wherein the listening module is configured to receive an alert signal from a user, and wherein the alerting module is configured to issue an alert responsive to receipt of the alert signal by the listening module.
